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Intelligent Document Assistant: A Retrieval-Augmented Generation (RAG) System

Project ID: PRJ073

Team ID: T073

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Project Trainer / Guide

Mr. SHIRWIN PRINCE I

Table of Contents

1.Introduction Section 1

2.Problem Statement & Objective Section 2

3.System Architecture(RAG pipeline) Section 3

4.Implementation Details & Tech Stack Section 4

5.Result & Testing Section 5

6.Conclusion & Future Scope Section 6

1. Introduction

Creative writing plays an important role in storytelling, entertainment, and communication. However, many people struggle to transform their imagination and simple ideas into meaningful stories due to a lack of inspiration, creativity, and proper story structure. Artificial Intelligence (AI), particularly Large Language Models (LLMs), provides new opportunities to simplify the creative writing process by understanding user inputs and generating meaningful text. StorySpark AI is designed as an AI-powered story and script idea generation platform that helps users convert their basic ideas into engaging story outlines and creative scripts. The platform allows users to explore different genres, such as Fantasy, Adventure, Romance, Mystery, Comedy, and Horror, and develop stories based on their preferences. By using an LLM-based approach, with Google Gemini proposed as the AI model, the system aims to generate relevant story ideas, characters, plots, and creative content based on user prompts. The project also provides features for managing user profiles and saving previously generated stories. The main objective of StorySpark AI is to encourage creativity, reduce the difficulties involved in story development, and provide an accessible environment for aspiring writers, students, and creative individuals to bring their imagination to life.

2. Problem Statement & Objective

Many people have creative ideas and a strong interest in storytelling but face difficulties in converting their thoughts into well-structured and engaging stories. Developing a complete story requires creativity, proper planning, character development, and a clear storyline. Beginners and aspiring writers often experience a lack of inspiration, writer's block, and difficulty in organizing their ideas. Traditional writing methods can also be time-consuming and may limit the exploration of different story possibilities. Therefore, there is a need for a simple and user-friendly platform that helps users transform their

basic ideas into meaningful stories and scripts. StorySpark AI aims to address these challenges by providing an AI-powered creative writing environment that assists users in generating story ideas, developing characters, creating plots, and exploring different genres based on their preferences.

Objectives:

The main objectives of the StorySpark AI project are:

- To develop an AI-powered application that generates creative stories based on user-provided ideas.
- To provide different story genres such as Fantasy, Romance, Adventure, Mystery, Comedy, and Horror.
- To generate story outlines, characters, plots, and scripts based on user preferences.
- To use an AI model to generate meaningful and creative story content.
- To provide a user-friendly platform for creating and managing stories.
- To implement secure user login and registration using Firebase Authentication.
- To store user profiles and generated stories using Firebase Firestore.
- To help users overcome writer's block and improve their creativity.

3. System Architecture (RAG pipeline)

StorySpark AI can use Retrieval-Augmented Generation (RAG) to improve the quality and consistency of AI-generated stories. RAG combines information retrieval with a Large Language Model (LLM). It retrieves relevant information from previously stored stories and provides that information to the AI model before generating new content. The RAG pipeline consists of two main processes: the document ingestion process and the query processing process.

1.1 Data Collection Phase

Previously saved stories, character descriptions, plot details, and other relevant story information are collected from the database. This information is used as a knowledge source for future story generation. It helps the system understand the user's previous creative work and maintain the required story details.

1.2 Text Extraction and Chunking Phase

The collected story information is extracted and divided into smaller, meaningful sections called chunks. Each chunk contains a specific part of the story, such as character descriptions, story backgrounds, or important events. This makes the information easier to process, search, and retrieve.

1.3 Embedding Generation Phase

The text chunks are converted into numerical representations called embeddings using an embedding model. These embeddings capture the meaning and context of the text. This allows the system to identify similar story information even when different words are used in the user's query.

1.4 Vector Database Storage Phase

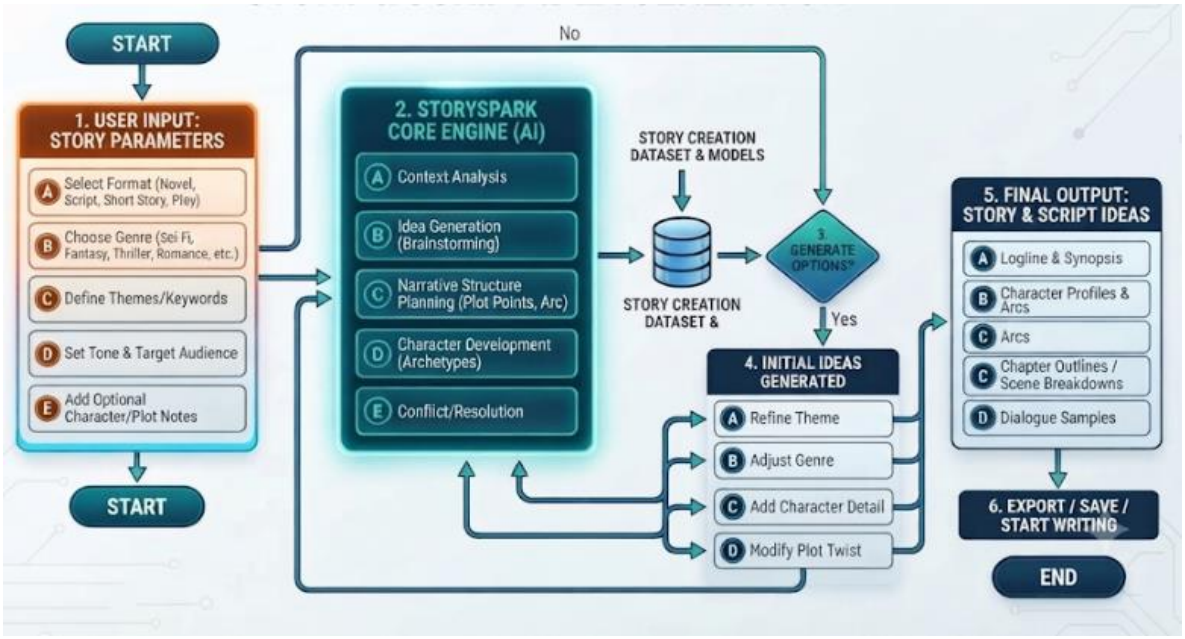
The generated embeddings and their corresponding original text chunks are stored in a Vector Database. The database maintains the information so that it can be searched and retrieved whenever the user requests a new story or wants to continue a previously generated story.

1.5 Query and Retrieval Phase

When a user submits a prompt, the query is converted into an embedding using the same embedding model. The Vector Database performs a similarity search to identify the most relevant stored story chunks. The retrieved information may include character details, previous events, and storyline information related to the user's request.

1.6 LLM Response Generation Phase

The retrieved story chunks are combined with the user's prompt and provided as context to the Large Language Model (LLM), such as Google Gemini. The LLM processes the input and generates a meaningful and relevant story response. The generated content is then displayed to the user through the StorySpark AI application. This helps maintain character consistency, improve storyline continuity, and reduce unrelated or incorrect responses.



4. Implementationa Details & Tech Stack

Component	Technology Used	Purpose
User Registration	React.js,Firebase Authentication	Allows new users to create an account using email and password.
User Profile	React.js, Firebase Authentication	Authenticates registered users and provides secure access
User Login	React.js, CSS, Cloud Firestore	Allows users to enter, view, and update their personal details and preferences.
Dashboard	React.js, HTML, CSS	Provides a central interface to access application features.
Story Idea Input	React.js, JavaScript	Allows users to enter their basic story ideas.
Genre Selection	React.js, JavaScript	Allows users to choose genres such as Fantasy, Romance, Comedy, Adventure, and Mystery.
AI Story Generation	Google Gemini API (if integrated), JavaScript	Generates creative story content based on the user's input and selected genre.
Story Display	React.js, HTML, CSS	Displays the generated story in a readable format.
Story Storage	Cloud Firestore	Stores user profiles and generated story information.
Frontend Development	React.js, HTML, CSS, JavaScript	Develops the application's user interface and interactive features.
Build and Development	Vite	Runs the development server and builds the application for deployment.
Development Environment	Visual Studio Code	Used to write, edit, and run the application.
Version Control	Git, GitHub	Maintains and manages project source code.
Deployment	Render	Hosts the application online.

5. Result & Testing

The StorySpark AI project provides a simple and user-friendly platform that helps users develop their creative story ideas. Users can register and log in to the application, access the dashboard, manage their profile details, enter their basic story ideas, and select their preferred genres. The application aims to make storytelling easier by helping users transform their ideas into meaningful story content. Testing is carried out to verify the functionality of different modules, including user registration, login, profile management, dashboard navigation, story idea input, and genre selection. The application is also checked for proper input handling, error messages, and smooth navigation between pages. If AI story generation is integrated, it is tested to ensure that the generated content matches the user's input and selected genre. Overall, testing helps identify errors, improve application reliability, and provide a better user experience.

 Unit Testing (Core Engine): Evaluated individual modules (Context Analysis, Theme Extraction, and Narrative Structure Planning) to ensure parameter mapping worked accurately.

 Integration Testing: Tested the pipeline flow from **User Input (Step 1)** through the **AI Core Engine (Step 2)** to the **Dataset/Model Retrieval (Step 3)** to confirm zero data loss and smooth state transitions.

 Prompt Robustness & Edge Cases: Tested extreme or ambiguous user parameters (e.g., conflicting genres like "Romantic Horror-Comedy") to evaluate how gracefully the AI handles contradiction.

 User Acceptance Testing (UAT): Conducted trials with beta users (writers and content creators) to assess the creative viability, tone consistency, and usefulness of the generated loglines, character arcs, and script outlines.

6. Conclusion & Future Scope

StorySpark AI is designed to provide a simple and creative platform for users to develop their story ideas. The application allows users to register, log in, manage their profiles, enter story ideas, and select different genres through a user-friendly interface. By combining frontend technologies with Firebase services and AI-based story generation (where integrated), the project aims to make storytelling easier and more accessible. It helps beginners, students, and aspiring writers explore their creativity and overcome difficulties in developing stories. Overall, StorySpark AI provides a useful approach to improving the creative writing experience through technology.

Fuure Scope

In the future, StorySpark AI can be enhanced by introducing additional features to improve its functionality and user experience. More story genres and languages can be added to support a wider range of users. The application can include advanced AI features for generating complete stories, dialogues, characters, and scripts. To expand the capabilities and market readiness of StorySpark AI, several key developments are planned for future phases:

- Real-Time Collaborative Writing Spaces
- Industry-Standard Script Formatting
- Advanced Style Customization & Fine-Tuning
- Expanded Multimedia & Storyboarding Integration